

Nine Questions That Make a Lesson Answerable

A teacher's guide to the Variation-Theory question sequence — what each question does, and how to write your own

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What this is

These are the recurring questions built into my biology slides. They are not decoration and they are not review; each one has a specific job in helping students learn to see an idea. This guide names each question type, explains what it does and the research it rests on, and gives you a prompt you can paste into your own AI assistant to write that kind of question for your own lessons.

The sequence comes from Variation Theory (Marton & Booth, 1997; Marton, 2015). Its central claim is plain: a student learns a concept only when they can discern its critical aspect — the one feature that separates this idea from the ones next to it — and a feature becomes discernible only when it is varied against a background that stays the same. You cannot notice “warm-blooded” until you have seen it change while other things hold still. The questions below are that logic turned into slides: name the feature, vary it, have students state the pattern, then break the pattern so the understanding has to hold up.

Why students write on the slides

There are no separate worksheets. Students annotate directly on the same slides I teach from, in the writing space built into each one. That choice is deliberate. What ends up captured is the student's own reasoning — the sentence they wrote to explain why — not a clean copy of my notes. Making student thinking visible, in the moment, is the core of formative assessment (Black & Wiliam, 1998): you cannot respond to reasoning you cannot see. Writing on the slide turns every question into a small, readable record of where a student actually is.

The nine question types

Each entry gives the plain job of the question, the term for it, the research behind it, and a prompt you can adapt.

1. Reconnect to the last lesson — the *Continuity question*

What it does. Opens the lesson by asking students to use yesterday's idea as the footing for today's, so the unit reads as one storyline instead of a pile of separate topics. This is the spiral in practice — returning to an idea and carrying it one step further (Bruner, 1960).

Example prompt

Write a CONTINUITY QUESTION to open a lesson. I'll give you the previous lesson's main idea and today's main idea. Write one question, plain language, that can only be answered using yesterday's idea and that leads into today's. One or two sentences, works as a discussion opener, not a quiz.

2. Open with an approachable hook — the *Bellringer*

What it does. A low-stakes question that surfaces what students already know and gets them thinking about the topic before instruction. Prior knowledge is the ground everything else is built on — “the most important single factor influencing learning is what the learner already knows” (Ausubel, 1968, p. vi). The hook is meant to be reachable by everyone, not hard.

Example prompt

Write a **BELLRINGER** question for a 9th-grade biology lesson on [topic]. It should be approachable, spark curiosity, and pull up what students may already know. Everyone should be able to write something. One sentence, plus room to answer.

3. Name the one feature to notice — the *Critical Aspect*

What it does. Puts the single feature students must learn to discern in front of them, as a narrow question, without giving the answer. Its whole job is to fix attention on the object of learning and nothing else (Marton, 2015). One aspect per question — not two.

Example prompt

I'm teaching [concept]. Name the ONE critical aspect — the single feature that distinguishes it from nearby ideas — and write one narrow question that points students at just that feature without revealing the answer. Plain language, answerable in 1–2 minutes.

4. Put two cases side by side — the *Contrast Set*

What it does. Shows two organisms or cases that differ in that one feature, in two columns, with a question that forces a comparison. This is the variation that makes the aspect discernible: you learn what something is by seeing what it is not (Marton, 2015). The rest is held constant so the one difference stands out.

Example prompt

For the critical aspect [aspect], give me a CONTRAST SET: two organisms or cases that differ in that feature and are otherwise similar. Lay them out as two columns and end with one question that forces students to compare them on that feature.

5. Turn the pattern into a rule — the *Build a Rule*

What it does. Has students put the pattern they just saw into words by finishing a sentence frame: “Finish this sentence as a rule: ‘___ works when ___.’” This is generalization, and it works because students construct the rule themselves rather than receive it (Wittrock, 1974). Saying the rule is how the aspect gets separated from the noise around it.

Example prompt

After a contrast on [aspect], write a BUILD A RULE slide: a sentence frame students finish to state the rule in their own words, in the form “[concept] works/happens when ____.” One blank, sized for a one- to two-line answer.

6. Break the rule on purpose — the *Pattern Break*

What it does. Presents the case that breaks the too-simple rule students just wrote, forcing them to revise it. The productive discomfort of meeting an example your rule cannot handle is the engine of conceptual change (Posner, Strike, Hewson, & Gertzog, 1982). The breaking case should come from the same family as the contrast, not out of nowhere.

Example prompt

Students just built the rule “[rule].” Give me a PATTERN BREAK: a real organism or case that fits the topic but breaks that too-simple rule, plus one question that makes students notice the break and revise the rule.

7. Ask it at three depths — the *3-Tier Concept Question*

What it does. Asks the same critical aspect at three self-selected depths — Getting Started, Working On It, Mastery — so every student can enter and stretch on one idea. The three are scaffolded depths of a single question, not three different questions, which lets each student work at the edge of what they can currently do (Vygotsky, 1978; Tomlinson, 2014). Students choose their tier; the slide never assigns it. All three prompts stay uniformly plain to read — the difficulty lives in the biology, never in the sentence — because tangled wording measures reading, not the concept (Abedi & Lord, 2001).

Example prompt

Write a 3-TIER CONCEPT QUESTION for the critical aspect [aspect].

- *Getting Started*: concrete recall or identify-an-example, answerable from memory.
- *Working On It*: apply the rule; ask WHY a simple version isn't enough.
- *Mastery*: explain the mechanism; ask WHAT SPECIFICALLY the rule must rest on.

Keep the three labels exactly as written and in that order; all three address the same aspect. Plain language at a 7th-grade reading level. Add a word bank of 4–6 single words in [brackets]. End with: “Think first. Discuss with a partner. Then write.”

8. Push it into a “what if” — the *What-If*

What it does. A counterfactual that cannot be factually wrong, where students reason forward about consequences and carry the aspect into a new situation. This is where an idea is tested for transfer — whether it holds when the context changes (Perkins & Salomon, 1988). The evolution and ecology versions are the strongest, because a changed condition ripples in ways students can reason through.

Example prompt

Write a WHAT-IF question for [concept]. Pose a counterfactual (“What if [condition] changed?”) that can't be factually wrong and makes students reason about the consequences using [aspect]. Favor an evolution or ecology framing if the topic allows.

9. Relate it to your own life — the *Relate-to-Me*

What it does. A closing reflection: “How ____ relates to me is ____.” Having students write, in their own words, how material connects to their lives raises both interest and achievement, with the largest gains for the students who were struggling most (Hulleman & Harackiewicz, 2009). It is not a warm add-on; it is one of the better-evidenced moves in the sequence, and it fits a Title I and multilingual classroom especially well.

Example prompt

Write a RELATE-TO-ME closing prompt for [concept]. Ask students to write, in their own words, how this idea connects to their own life, choices, or community — in the frame “How ____ relates to me is ____.” Open-ended; there is no wrong answer.

The conditional model: what builds up and drains away — the *Stock-and-Flow (Tank-and-Fill)*

Some ideas are not a feature to discern but a quantity that accumulates in a store and drains through flows: energy across trophic levels, carbon or water moving through a cycle, a population rising and falling. Where a lesson carries one of these, a stock-and-flow model — a labeled picture of the tank and its flows, plus a control students can vary — makes the behavior visible in a way a static diagram cannot (Meadows, 2008). This one is conditional: you use it only where the content actually has a stock and a flow in it, never forced onto a lesson that has neither.

In my store, these are the small interactive models offered free, each linked to the specific critical aspect it illustrates.

Using this with your own slides

You do not need all nine in every lesson. The five in the middle — critical aspect, contrast, build a rule, pattern break, three-tier — are the core arc, and they repeat once per critical aspect in a lesson. Continuity and the bellringer open; what-if and relate-to-me close; the stock-and-flow appears only where the content earns it. Start by naming the one critical aspect of your lesson, and let that choice drive the rest — every question above is written around that single feature.

A companion map, *The Year in Critical Aspects*, lists every critical aspect students meet across the whole biology year, so you can see where any one lesson sits in the larger storyline.

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